

Binary Detection

June 16, 2026

1 Class Assignments

1.1 Assignment 4: MAP Detector versus Machine-Learning Detector

Objective. Compare classical statistical detection with supervised machine-learning detectors for the binary AWGN detection problem.

Problem statement. Use the same binary waveform model:

$$r_k[n] = a_k + \eta_k[n], \quad n = 0, 1, \dots, L-1, \quad (1)$$

where

$$a_k \in \{a_0, a_1\}, \quad \eta_k[n] \sim \mathcal{N}(0, \sigma_\eta^2). \quad (2)$$

Compare analytical MAP detection against machine-learning classifiers trained from simulated labeled data.

Tasks.

1. Generate a labeled dataset of received symbol blocks:

$$\{(\mathbf{r}_k, b_k)\}_{k=1}^N, \quad (3)$$

where $\mathbf{r}_k \in \mathbb{R}^L$ and $b_k \in \{0, 1\}$.

2. Split the dataset into training, validation, and test sets.
3. Implement the analytical scalar MAP detector using one sample per symbol.
4. Implement the analytical matched-filter MAP detector using

$$Z_k = \frac{1}{L} \sum_{n=0}^{L-1} r_k[n]. \quad (4)$$

5. Train a logistic-regression classifier using the matched-filter statistic Z_k as input.
6. Train a multilayer perceptron (MLP) using the matched-filter statistic Z_k as input.
7. Train an MLP using the full received block $\mathbf{r}_k$ as input.
8. Compare all detectors using:

$$P_e, \quad P_{\text{FA}}, \quad P_{\text{M}}, \quad P_{\text{D}} = 1 - P_{\text{M}}. \quad (5)$$

9. Discuss whether the machine-learning detectors improve upon the analytical MAP detector.

Required deliverables.

1. Description of the generated training and test datasets.
2. Description of each detector and classifier.
3. Performance table of all detectors:

Detector	Input	P_e	P_{FA}	P_{M}
Scalar MAP	One sample			
Matched-filter MAP	Z_k			
Logistic regression	Z_k			
MLP	Z_k			
MLP	$\mathbf{r}_k$			

4. Confusion matrix for each detector.
5. Short discussion addressing the following question:

When does supervised learning reproduce the analytical detector, and when does it fail to improve over it?